

Quiz Master App-V1

Project by [Aanchal Routela](#)

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MODERN APPLICATION DEVELOPMENT – 2

May, 2025 Term

Student Details:-

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Project Description:-

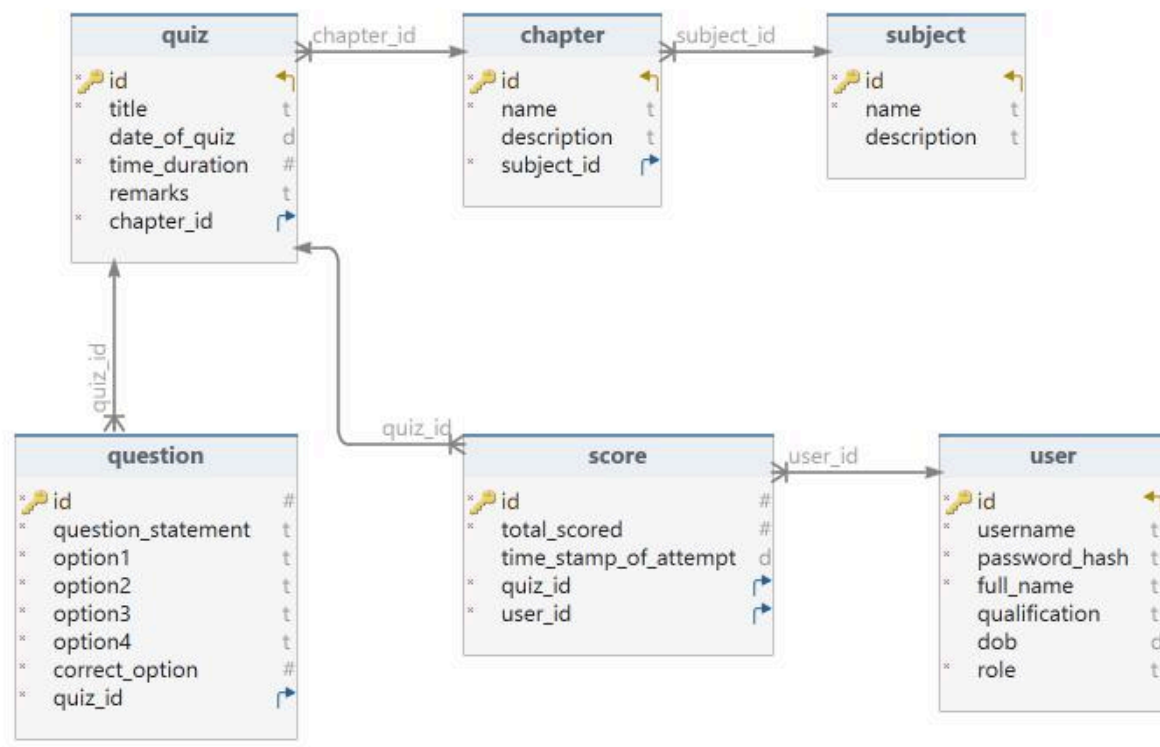
This multi-user application serves as a comprehensive exam preparation platform for various courses. It features an administrator role to create subjects, chapters, quizzes and manage user accounts, while other users can give quizzes and practice for the exam. A key feature is the timed quiz function, which simulates real exam conditions and helps users manage their time effectively.

TECHNOLOGIES Used:-

- ☐ **Flask:** Lightweight Python web framework to handle the backend API.
- ☐ **Flask-RESTful:** For structuring and creating clean RESTful APIs.
- ☐ **Flask-SQLAlchemy:** ORM for defining models and managing database operations easily.
- ☐ **Flask-Migrate:** For handling database migrations using Alembic.
- ☐ **Flask-JWT-Extended:** For secure token-based user authentication and authorization.
- ☐ **Flask-CORS:** To enable Cross-Origin Resource Sharing between frontend and backend.
- ☐ **Flask-Mail:** For sending emails like quiz notifications or account confirmations.
- ☐ **SQLite:** Lightweight embedded database used for development (can be switched to PostgreSQL/MySQL in production).
- ☐ **Vue.js:** JavaScript framework used for building interactive user interfaces.
- ☐ **Vue-Router:** Handles frontend routing for navigating between different pages.
- ☐ **Axios:** For making HTTP requests from Vue frontend to Flask backend.
- ☐ **Bootstrap:** CSS framework to design responsive and styled UI components.
- ☐ **Chart.js:** For creating visually appealing charts and graphs in user/admin dashboard.

- ☐ **JWT (JSON Web Tokens)**: Used for managing sessions and secure communication between frontend and backend.
- ☐ **Role-Based Access Control (RBAC)**: Used to differentiate permissions between users and admins.

DB SCHEMA DESIGN:-



API Endpoints :-

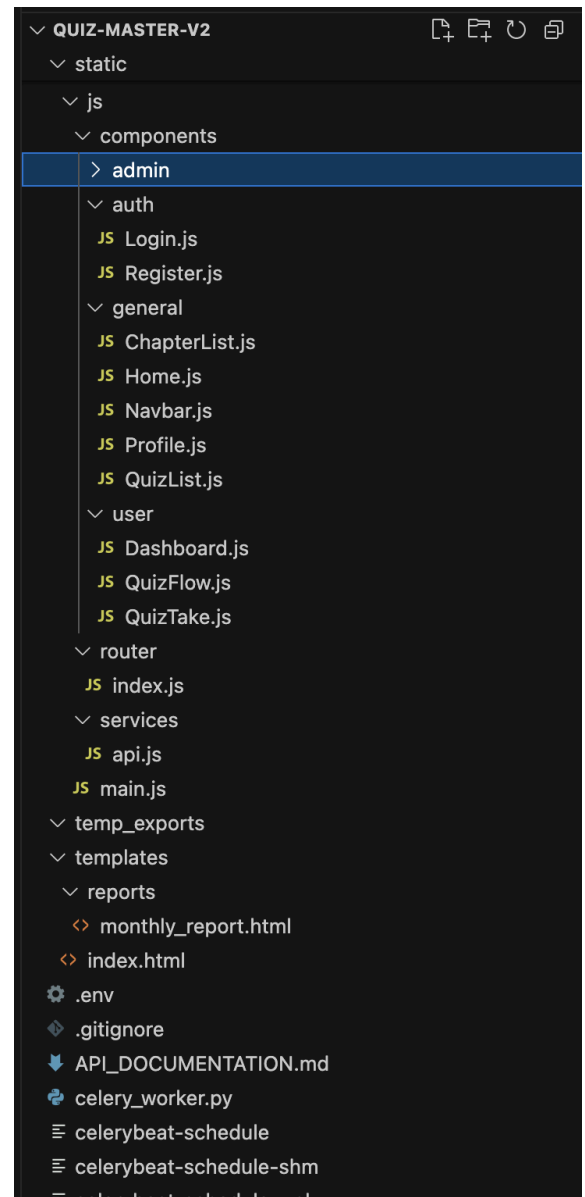
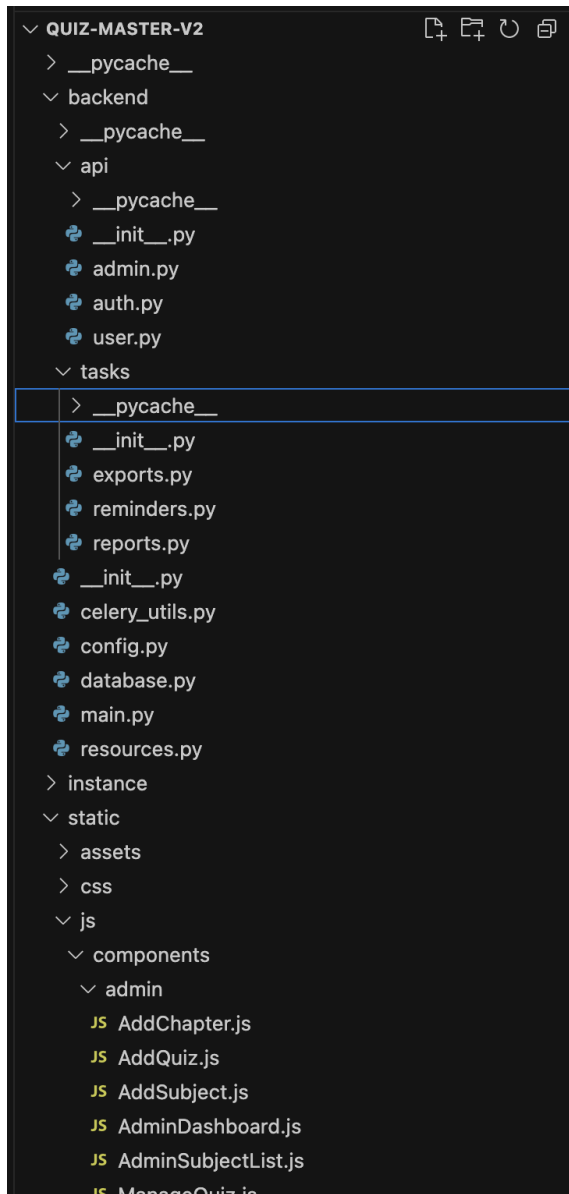
```

api.add_resource(UserApi, "/api/user", "/api/user/<int:id>")
api.add_resource(QuizApi, "/api/quiz", "/api/quiz/<int:id>")
api.add_resource(QuestionApi, "/api/question", "/api/question/<int:id>")
api.add_resource(ResultApi, "/api/result", "/api/result/<int:id>")
  
```

VIDEO LINK:-

[Click here to view video introduction](#)

Directory Structure:-



Usage of AI:-

AI/LLM was used for about 12% of the project, mainly for code suggestions, debugging, styling tips, and some help with export page design and Celery background task setup.